

3.5.4 Atoms and nuclear radiation

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 10			
Some atomic nuclei are unstable. The nucleus gives out ionising radiation. This is a random process called radioactive decay. The nuclear radiation emitted may be: • alpha particles • beta particles • gamma rays. Properties of alpha particles, beta particles and gamma rays limited to their penetration through materials and their range in air. Students will be expected to know some of the uses and dangers of the three types of radiation.		6.4.2.1	4.3.2.2